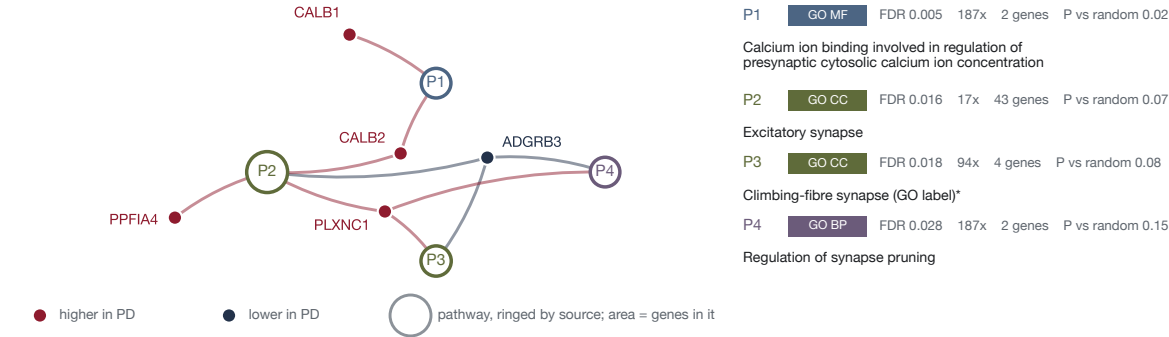


Source	Pathway	Genes in term / panel	Fold enrichment	-log ₁₀ FDR	FDR	P vs random lists	BCL11A	CALB1	CALB2	DNAJB1	FUS	PLXNC1	PPFIA4	TRRAP	ZMYND8	ADGRB3	SH3GL3	SOX6
1 significant	Higher in PD (18 genes)																	
GO MF	Calcium ion binding involved in regulation of presynaptic cytosolic calcium ion concentration	2 / 2	312x		0.002	0.01												
GO CC	Excitatory synapse	43 / 3	22x		0.058	0.41												
GO MF	Transcription coregulator activity	241 / 5	6x		0.058	0.41												
0 significant	Lower in PD (12 genes)																	
WikiPW	15q25 copy number variation	18 / 1	26x		0.060	0.62												
WikiPW	Oligodendrocyte development	15 / 1	31x		0.060	0.62												
WikiPW	10q22q23 copy number variation	19 / 1	25x		0.060	0.62												
4 significant	All 30 panel genes																	
GO MF	Calcium ion binding involved in regulation of presynaptic cytosolic calcium ion concentration	2 / 2	187x		0.005	0.02												
GO CC	Excitatory synapse	43 / 4	17x		0.016	0.07												
GO CC	Climbing-fibre synapse (GO label)*	4 / 2	94x		0.018	0.08												
GO BP	Regulation of synapse pruning	2 / 2	187x		0.028	0.15												

Shaded rows pass FDR 0.05. Fold enrichment: panel genes in the term against what chance gives. "P vs random lists": how often 100 random lists of the same size reached a smaller FDR.
*GO "cerebellar climbing fibre to Purkinje cell synapse": a label for a type of excitatory synapse, named after the tissue where these genes were first described - not a midbrain structure.

b Every significant pathway and the panel genes that carry it



Four pathways rest on five genes, three of them shared. Those genes are largely markers of which dopamine neurons survive, so part of this signal is the neuron mix rather than a pathway (Figure 10).

c The same settings on 100 random gene lists of the same size

